Options I: Introduction

BUSS386. Futures and Options

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Lecture Outline

- What an option is — and why only one side of the contract is bound
- The three reasons they exist: insurance, leverage, and income
- Payoff and profit for all four positions, and where the kink comes from
- Moneyness, intrinsic value, and what you are really buying with time value
- How to read a real option chain, and which options can be exercised early
- Where the volume actually is — and where options hide in plain sight

Derivatives Review — Where Do Options Fit?

Contract	Buyer's stance	Seller's stance
Forward / Futures	Obligation	Obligation
Swap	Obligation	Obligation
Option	**Right (chooses)**	**Obligation (follows buyer)**

- In futures/forwards/swaps both sides are bound.
- In options, only the **seller** is bound. The buyer pays an upfront *premium* for the right to walk away.
- This asymmetry $\Rightarrow$ option payoffs are *non-linear* (kinked at the strike). Everything that follows comes from this.

The first non-linear payoff

Everything in Lectures 3–5 was *linear*: a forward, a future, a swap all move one-for-one with the underlying. An option does not. Same stock, same \$100 strike, three positions:

S_T	Hold the stock	Long forward @ \$100	Long call @ \$100 ($c = \$5$)
\$80	\$80	$-$**\$20**	$-$**\$5**
\$100	\$100	\$0	$-$\$5
\$120	\$120	$+$**\$20**	$+$**\$15**

- The forward column is a **straight line**. The call column **bends** at \$100.
- What the \$5 premium buys: the $-$\$20 becomes $-$\$5. You keep the upside and truncate the downside.
- That bend is the whole subject. Pricing, the Greeks, and every strategy in Lectures 7–13 exist because of it.

What an Option Is

What Is an Option?

Def. An **option** is a contract that gives the holder the **right** (but not the obligation) to **buy** (call) or **sell** (put) an underlying asset at a fixed price on or before a fixed date.

- Building blocks:
  - Underlying S_t — stock, index, FX, futures, commodity, crypto
  - Strike price K — the promised trade price
  - Expiration T — last date the right can be used
  - Premium c or p — price paid up front
  - Style: **European** (exercise only at T) vs **American** (anytime up to T)
- Long vs short:
  - **Long** = buy the option (pay premium, hold the right)
  - **Short** = sell/write the option (collect premium, take the obligation)

Why Do Options Exist? — Three Uses

1. **Insurance / Hedging.**
   A long put on a stock you own is portfolio insurance: caps the downside. Cost = premium = the "deductible."

2. **Leverage / Speculation.**
   \$1,000 in MSFT stock at \$500 $\Rightarrow$ 2 shares.
   \$1,000 in MSFT \$510 calls at \$10 $\Rightarrow$ 100 shares of exposure.
   Same dollars in; very different P&L distribution.

3. **Income generation.**
   Covered call writing: sell calls on stock you own to collect premium in exchange for capping upside. Popular with pension and ETF strategies (e.g., JEPI, QYLD).

Example — Long Call on KOSPI 200

Setup. You buy a December 2026 KOSPI 200 European call, strike $K = 1{,}100$, premium $c = 25$ index points. Multiplier = ₩250,000 per point.

Q1. Index closes at $S_T = 1{,}160$ on expiry. Do you exercise? Payoff and profit?

A1. $S_T > K$, so exercise.
Payoff $= \max(1{,}160 - 1{,}100,\ 0) = \boxed{60}$ pts $=$ ₩15,000,000.
Profit (ignoring TVM) $= 60 - 25 = \boxed{35}$ pts $=$ ₩8,750,000.

Q2. Index closes at $S_T = 1{,}050$. Do you exercise?

A2. $S_T < K$, do not exercise.
Payoff $= 0$; Profit $= -25$ pts $= \boxed{-₩6{,}250{,}000}$.

Example — Long Put on Samsung Electronics

Setup. You buy a 3-month put on 005930.KS, $K = ₩280{,}000$, premium $p = ₩3{,}000$ per share. (KRX equity options carry a multiplier of **10 shares**; we work per share below, so multiply by 10 for one contract.)

Q1. Samsung closes at ₩272,000 on expiry. Exercise? P&L per share?

A1. $S_T < K$, so exercise.
Payoff $= \max(280{,}000 - 272{,}000, 0) = ₩8{,}000$.
Profit $= 8{,}000 - 3{,}000 = \boxed{₩5{,}000}$ per share.

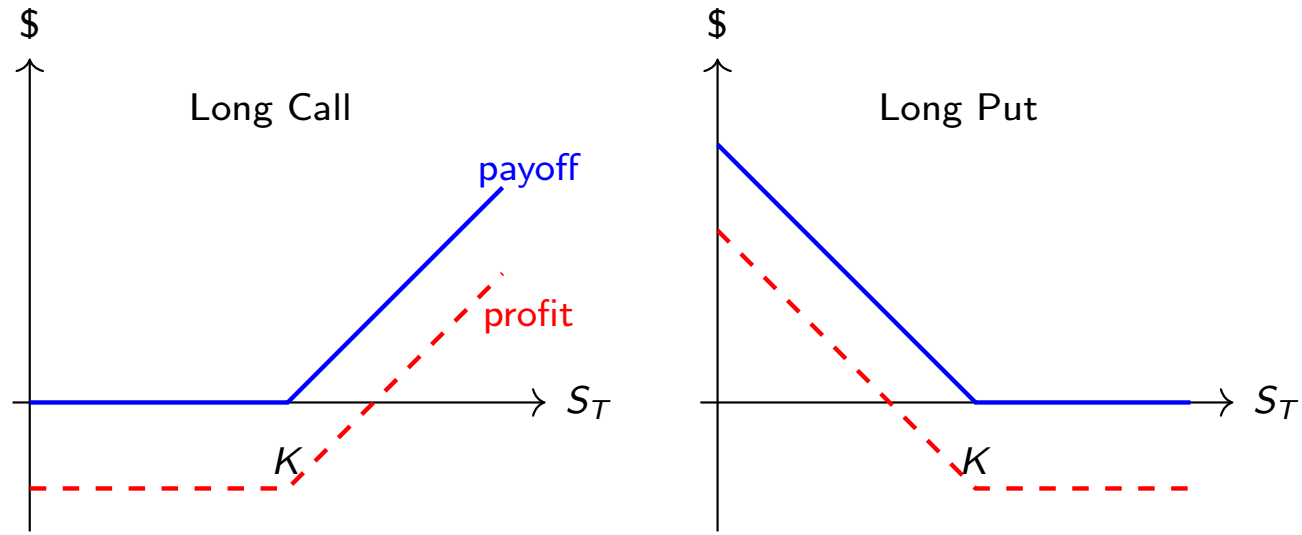
Q2. Samsung closes at ₩285,000. Exercise? P&L?

A2. Do not exercise.
Loss = premium paid $= \boxed{-₩3{,}000}$ per share.

Payoff and Profit

General Payoff — Long Call and Long Put

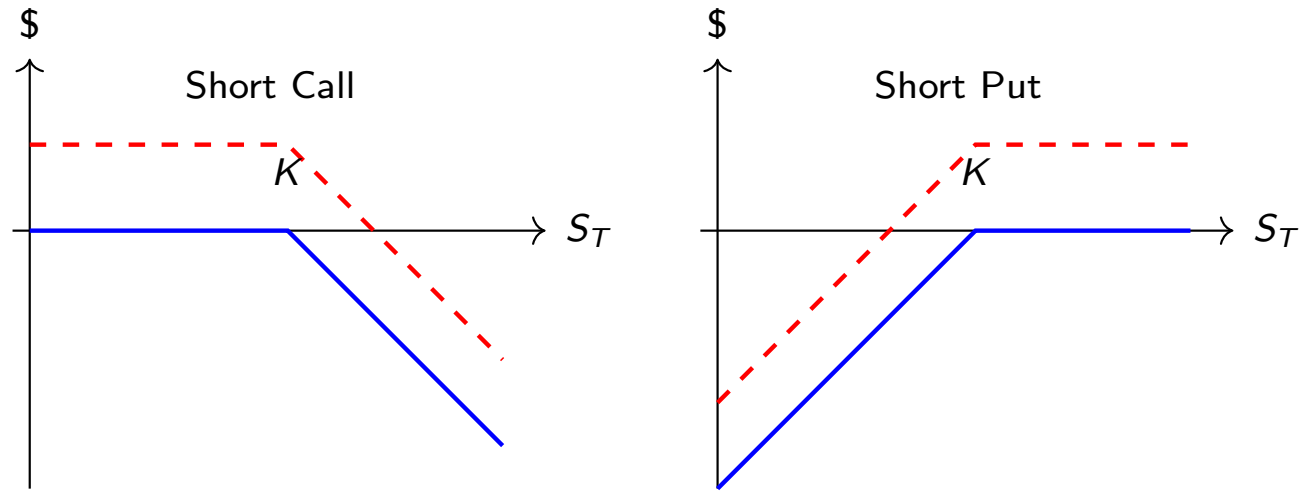
Long call: payoff $= \max(S_T - K, 0)$, profit = payoff $-c$.
Long put: payoff $= \max(K - S_T, 0)$, profit = payoff $-p$.



- Payoff curves are **kinked at** K — the source of non-linearity.
- Profit = payoff shifted down by the premium (no TVM adjustment by convention).

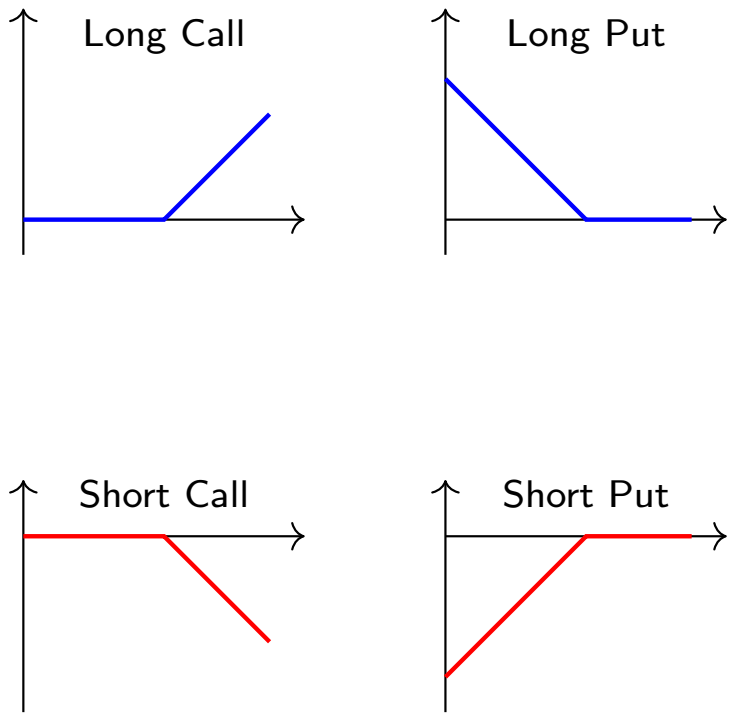
Short Positions — Mirror Images

Short call: payoff $= -\max(S_T - K, 0)$
Short put: payoff $= -\max(K - S_T, 0)$



- Seller's max gain = premium collected; max loss = (large or unbounded).
- Short call has **unlimited** downside if $S_T \to \infty$.
- Short put loss bounded by $K - 0 = K$ (stock can't go below zero).

The Four Fundamental Positions — One Picture



- Every option position — and every option *strategy* (spreads, collars, straddles) — is a linear combination of these four.
- Lecture 7 will build strategies by stacking these.

Reading Option Prices — and How the Market Works

Moneyness

	Call	Put
$S_t > K$	in-the-money (ITM)	out-of-the-money (OTM)
$S_t = K$	at-the-money (ATM)	at-the-money (ATM)
$S_t < K$	out-of-the-money	in-the-money

- "In-the-money" means *would pay something if exercised right now.*
- ATM options have the most uncertainty about whether they will end ITM — they trade with the highest *time value* (next slide).
- Quote convention: "KOSPI 200 1,100 call" usually means $K = 1{,}100$ on the nearest standard expiry.

Intrinsic Value + Time Value

Decomposition:

$$\underbrace{\text{Option price}}_{\text{quoted}} = \underbrace{\text{Intrinsic value}}_{\max(S_t - K, 0) \text{ for call}} + \underbrace{\text{Time value}}_{\geq 0}$$

- Intrinsic value: payoff if exercised *now*.
- Time value: extra premium for the chance the option moves further ITM before expiry.
- Time value $\to 0$ as $t \to T$ ("theta decay").
- Example: MSFT at \$500, June \$490 call quoted \$15. Intrinsic $= 10$, time value $= 5$.

Reading an option chain

KOSPI 200, December 2026 expiry. Index at 1,105. Calls on the left, puts on the right, strikes down the middle.

CALLS					PUTS			
Bid	Ask	Vol	Open int	**Strike**	Bid	Ask	Vol	Open int
49.90	50.45	1,240	8,310	**1,075**	18.25	18.70	2,880	15,740
40.90	41.40	3,180	14,620	**1,090**	24.15	24.60	4,110	19,230
32.95	33.45	6,450	21,870	**1,105**	31.10	31.60	5,730	20,410
26.15	26.60	4,920	18,340	**1,120**	39.20	39.75	3,260	13,880
20.35	20.80	2,610	11,050	**1,135**	48.35	48.90	1,470	7,920

- **You pay the ask and receive the bid.** Buying the 1,105 call costs $33.45 \times$ ₩250,000 = ₩8,362,500. The 0.50-point spread is the market maker's.
- **Moneyness** Calls below 1,105 are ITM, puts above it are ITM — and the two sides are mirror images.
- **Volume is today; open interest is everything still outstanding**

Which Options Can Be Exercised Early?

	Exercise	Settles in
US single stocks	**American**	Shares (100)
US index (SPX)	**European**	Cash
All KRX listed options	**European**	Cash

- US Options Market
  - Combines American/Physical (single stocks) and European/Cash (index options).
  - Demands active monitoring of dividend dates, early exercise math, and assignment risk.
- Korea Exchange (KRX)
  - Every KRX option (KOSPI 200, KOSDAQ 150, single stock) is European.
  - Single-stock options settle in cash despite having share underlyings.
  - No Early Exercise

KOSPI 200 https://global.krx.co.kr/contents/GLB/02/0201/0201040202/GLB0201040202.jsp SPX https://ww2.cboe.com/tradable_products/sp_500/spx_options/specifications

Where the Volume Is — and Who Can Take It Away

Contract / market	2025 ADV	
All US listed options	61 m/day	15.2 bn in 2025: a record
SPX index options	3.9 m/day	of which **0DTE: 2.3 m (59%)**
VIX options	858 k/day	options on volatility itself
KOSPI 200 options (monthly)	∼80 k/day	once **#1 in the world**

- **2011.** KOSPI 200 options traded 3.67 bn contracts — 43% of all global equity-index futures and options.
- **2012.** KRX raised the multiplier to ₩500k to cool retail speculation. Volume more than halved.
- **2025.** India's SEBI pulled the same lever. Global listed-option volume fell 50% to 88.6 bn contracts — while US volume rose 26% to a record.

Sources: [Cboe 2025] [FIA 2026]

A Short History

- **1900**. Bachelier's PhD thesis prices options using Brownian motion. Ignored for 60 years.
- **1973**. **Black, Scholes, and Merton** publish the option-pricing formula. *Same year:* CBOE opens for trading.
- **1997**. Merton and Scholes win the Nobel Prize (Black died 1995).
- **1997**. **KRX launches KOSPI 200 options** — becomes the world's #1 listed option contract by volume for over a decade.
- **2012**. KRX raises KOSPI 200 option multiplier $5\times$(₩100k $\to$ ₩500k) to curb retail mania. Volume falls $>50\%$ month-on-month from June.
- **2017**. Multiplier cut to ₩250k to revive the market; volumes rebound.
- **2022–26**. **0DTE** options on SPX explode in volume.

Where Is the Option?

Options are rarely labelled "option." For each of these, name the **underlying**, the **strike**, and **who is short**.

1. Your offer letter grants you 1,000 shares at ₩50,000, vesting over four years.
2. A bond pays 1% a year and converts into 20 shares per $1,000 of face value.
3. A studio pays $5m today for the right, not the obligation, to make the sequel in 2029.
4. Your bank deposit is protected up to ₩50m if the bank fails.
5. A bank sells you a note paying 8% a year — unless the Hang Seng China Enterprises Index ever falls by half.

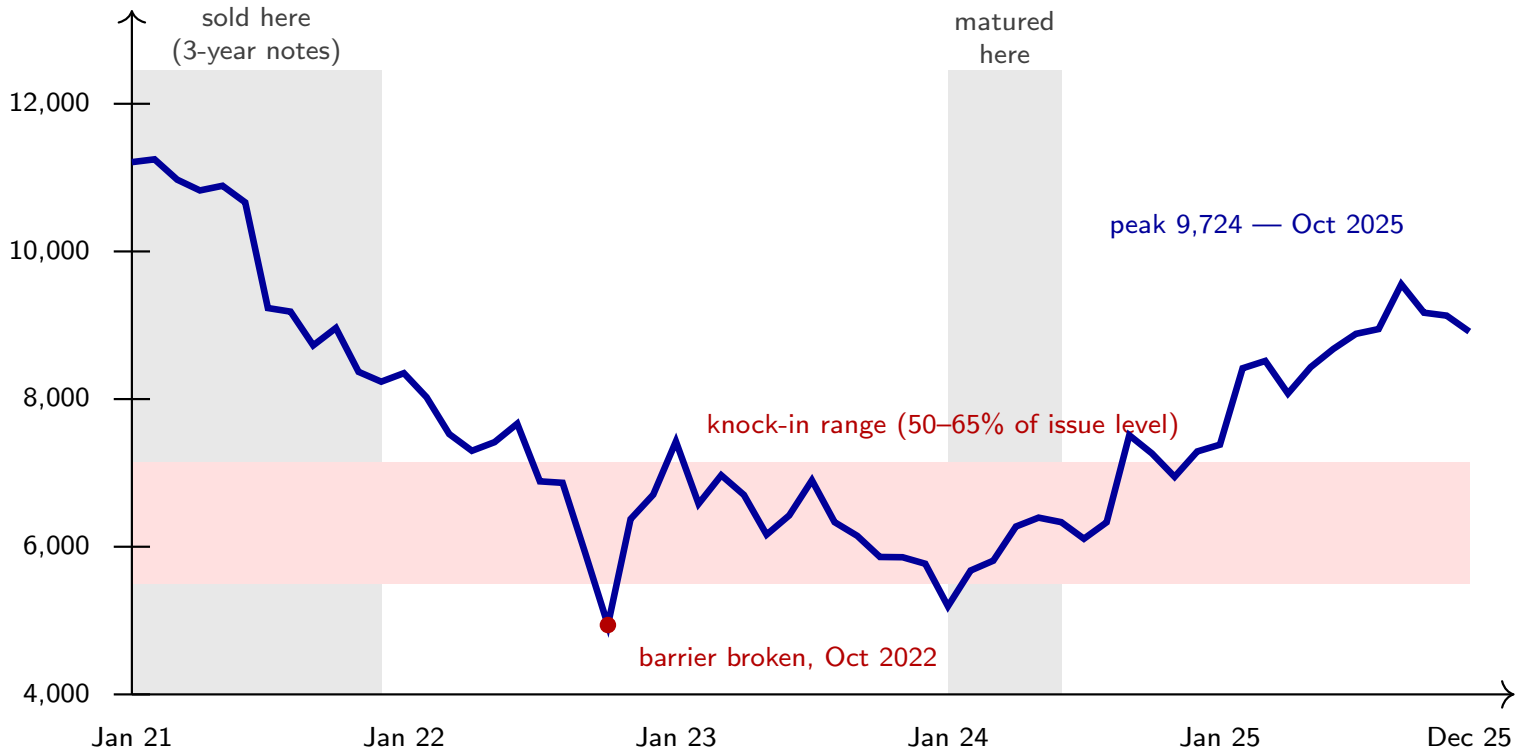
Case — Selling Puts to Depositors (Korea, 2024)

Korean banks and brokers sold **₩18.8 trillion** of ELS linked to the Hang Seng China Enterprises Index, largely to retail customers at bank branches.

- **The pitch.** A high fixed coupon "unless the index collapses," sold across the counter as a deposit substitute.
- **The instrument.** The buyer is **short a deep out-of-the-money put** on the HSCEI. The coupon *is* the premium.
- **How it pays.** Collect the coupon while the index stays above the **knock-in barrier** — typically 50–65% of the level at issue. Breach it, and at maturity the investor absorbs the index's full decline.
- **So the upside is capped and the downside is not.** Exactly the shape of a written put.

Source: [The Korea Herald, 2024–25]

Case — The HSCEI, 2021 to 2025



- The index fell through the knock-in range in late 2022 — 4,939 in October — long before any note matured.
- ~~₩~~4.6 tn ($3.2bn) realised; banks compensated over 96% and the FSS sanctioned five of them. The index bottomed at 5,002 on 22 January 2024, then reached 9,724 by October 2025 — up 94%, and far too late.

Data: Hang Seng China Enterprises Index, month-end closes [Yahoo Finance]

Exotic Options — Preview (Lecture 12)

- **Asian**: payoff depends on *average* price over a window. Used for FX hedging, oil swaps. Smoother than vanilla.
- **Barrier**: knocks in or out if the underlying crosses a level. Cheaper than vanilla; key building block of ELS.
- **Lookback**: payoff depends on max/min over the life.
- **Digital / Binary**: fixed payoff if a condition is met. Building block for autocallables.
- **Quanto**: payoff in a currency different from the underlying. Common in Korean-issued products linked to US indices.

Summary — what to take away

- An option binds **only the seller**. The buyer pays a premium for the right to walk away — and that asymmetry is where the kink comes from.
- The **four positions** — long and short, call and put — are the complete alphabet. Every strategy in this course is a combination of them.
- **Payoff** is what you receive at T; **profit** is payoff less the premium. Keep the two apart.
- Price = **intrinsic value** + **time value**, and time value decays to zero at expiry.
- Options hide in plain sight — employee grants, convertible bonds, deposit insurance, and the ELS sold to Korean savers as a deposit substitute.

Next week: what we can say about option prices *without* a pricing model — the six factors that move them, arbitrage bounds, put-call parity, and when early exercise is rational. Reading: Hull, Ch. 11.